Ist sessional Thermodynamics -Sept 2025

Q1. A certain amount of an ideal gas is initially at a pressure P_1 and temperature T_1 . First, it undergoes a constant pressure process 1–2 such that $T_2 = 3T_1 / 4$. Then, it undergoes a constant volume process 2-3 such that $T_3 = T_1/2$. . (i) draw the processes on PV diagram (ii)Find the ratio of the final volume to the initial volume of the ideal gas 05

Q.2 0.2 m^3 of an ideal gas at pressure of 2 MPa and 600 K is expanded isothermally to 5 times the initial volume. It is then cooled to 300 K at constant volume and then compressed back polytropically to its initial state. Determine the net work done and heat transfer during the cycle. 05

Q3. In a conference hall comfortable temperature conditions are maintained in winter by circulating hot water through a piping system. The water enters the piping system at 3 bar pressure and 50 °C temperature (enthalpy = 240 kJ/kg) and leaves at 2.5 bar pressure and 30 °C temperature (enthalpy = 195 kJ/kg). The exit from the piping system is 15 m above the entry. If 30 MJ/hr of heat needs to be supplied to the hall, make calculations for the quantity of water circulated through the pipe per minute. Assume that there are no pumps in the system and that the change in kinetic energy is negligible. 05